

### Getting Started

Start a new repo:  
git init

Clone an existing repo:  
git clone <url>

### Prepare to Commit

Add untracked file or unstaged changes:  
git add <file>

Add all untracked files and unstaged changes:  
git add .

Choose which parts of a file to stage:  
git add -p

Move file:  
git mv <old> <new>

Delete file:  
git rm <file>

Tell Git to forget about a file without deleting it:  
git rm --cached <file>

Unstage one file:  
git reset <file>

Unstage everything:  
git reset

Check what you added:  
git status

### Make Commits

Make a commit (and open text editor to write message):  
git commit

Make a commit:  
git commit -m 'message'

Commit all unstaged changes:  
git commit -am 'message'

### Move Between Branches

Switch branches:  
git switch <name>

OR  
git checkout <name>

Create a branch:  
git switch -c <name>

OR  
git checkout -b <name>

List branches:  
git branch

List branches by most recently committed to:  
git branch --sort=-committerdate

Delete a branch:  
git branch -d <name>

Force delete a branch:  
git branch -D <name>

### Diff Staged/Unstaged Changes

Diff all staged and unstaged changes:  
git diff HEAD

Diff just staged changes:  
git diff --staged

Diff just unstaged changes:  
git diff

### Diff Commits

Show diff between a commit and its parent:  
git show <commit>

Diff two commits:  
git diff <commit> <commit>

Diff one file since a commit:  
git diff <commit> <file>

Show a summary of a diff:  
git diff <commit> --stat  
git show <commit> --stat

### Ways to refer to a commit

Every time we say <commit>, you can use any of these:

|                   |                   |
|-------------------|-------------------|
| * a branch        | main              |
| * a tag           | v0.1              |
| * a commit ID     | 3e887ab           |
| * a remote branch | origin/main       |
| * current commit  | HEAD              |
| * 3 commits ago   | HEAD^^^ or HEAD~3 |

### Discard Your Changes

Delete unstaged changes to one file:  
git restore <file>

OR  
git checkout <file>

Delete all staged and unstaged changes to one file:  
git restore --staged --worktree <file>

OR  
git checkout HEAD <file>

Delete all staged and unstaged changes:  
git reset --hard

Delete untracked files:  
git clean

'Stash' all staged and unstaged changes:  
git stash

### Edit History

"Undo" the most recent commit (keep your working directory the same):  
git reset HEAD^

Squash the last 5 commits into one:  
git rebase -i HEAD~6

Then change "pick" to "fixup" for any commit you want to combine with the previous one

Undo a failed rebase:  
git reflog BRANCHNAME

Then manually find the right commit ID in the reflog, then run:  
git reset --hard <commit>

Change a commit message (or add a file you forgot):  
git commit --amend

### Code Archaeology

Look at a branch's history:  
git log main

git log --graph main  
git log --oneline

Show every commit that modified a file:  
git log <file>

Show every commit that modified a file, including before it was renamed:  
git log --follow <file>

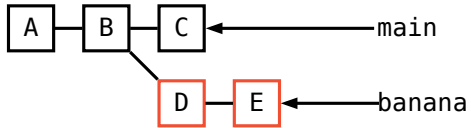
Find every commit that added or removed some text:  
git log -G banana

Show who last changed each line of a file:  
git blame <file>

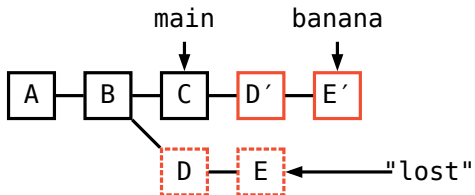
### Combine Diverged Branches

Combine with rebase:  
git switch banana  
git rebase main

Before:

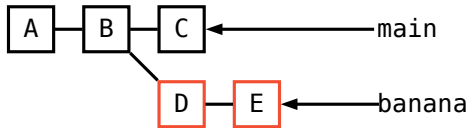


After:

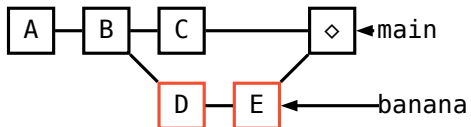


Combine with merge:  
git switch main  
git merge banana

Before:

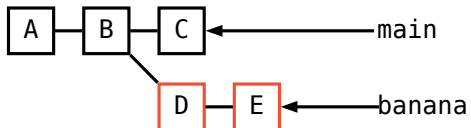


After:

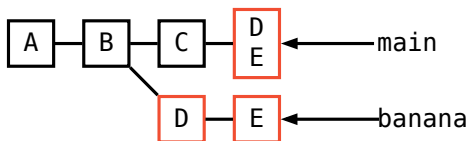


Combine with squash merge:  
git switch main  
git merge --squash banana  
git commit

Before:



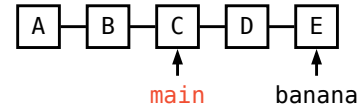
After:



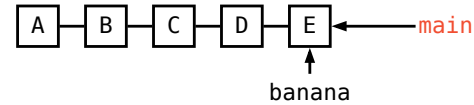
Bring a branch up to date with another branch  
(aka "fast-forward merge"):

git switch main  
git merge banana

Before:

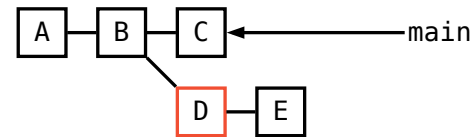


After:

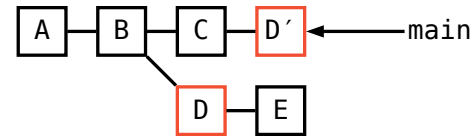


Copy one commit onto the current branch:  
git cherry-pick <commit>

Before:



After:



### Restore an Old File

Get the version of a file from another commit:  
git checkout <commit> <file>  
OR  
git restore <file> --source <commit>

### Add a Remote

git remote add <name> <url>

### Push Your Changes

Push the  
main

branch to the remote  
origin

:

git push origin main

Push the current branch to its remote "tracking  
branch":  
git push

Push a branch that you've never pushed before:  
git push -u origin <name>

Force push:

git push --force-with-lease

Push tags:

git push --tags

### Pull Changes

Fetch changes (but don't change any of your  
local branches):  
git fetch origin main

Fetch changes and then rebase your current  
branch:  
git pull --rebase

Fetch changes and then merge them into your  
current branch:  
git pull origin main

OR  
git pull

### Configure Git

Set a config option:

git config user.name 'Your Name'

Set option globally:

git config --global ...

Add an alias:

git config alias.st status

See all possible config options:

man git-config

### Important Files

Local git config:  
.git/config

Global git config:  
~/.gitconfig

List of files to ignore:  
.gitignore